

# Sulfate Injection and Ozone Recovery

How Stratospheric Aerosol Injection Could Delay the Healing of the Antarctic  
Ozone Layer

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## Introduction

Thanks to the Montreal Protocol, stratospheric chlorine has declined at roughly 0.8% per year since 1997, when it reached its peak (Strahan & Douglass, 2018). By 2016, Solomon et al. found significant healing of the Antarctic ozone layer, with ozone column amounts increasing at about 2.5 Dobson Units per year after 2000. Models project that the ozone hole will fully recover between 2060 and 2080, assuming no new perturbations (Newman et al., 2006).

Sulfate Aerosol Injection (SAI) injects  $\text{SO}_2$  into the lower stratosphere at approximately 20 km altitude to scatter incoming solar radiation (Tilmes et al., 2022). The idea draws on the 1991 eruption of Mount Pinatubo, which injected about 20 Tg of  $\text{SO}_2$  and cooled the planet by  $0.5^\circ\text{C}$  for two years (Portmann et al., 1996). Tilmes et al. (2008) noted, however, that Pinatubo enhanced ozone destruction in the polar regions in the years following the eruption.

This review determines the extent to which localized stratospheric heating from SAI alters the temperature difference between the tropics and poles, and how the resulting intensification of the Antarctic polar vortex alters the chemical kinetics of ozone depletion. The answer matters because SAI is increasingly modeled as a near-term climate intervention (Tilmes et al., 2022; Richter et al., 2017), and if it delays the ozone recovery promoted by the Montreal Protocol, that cost must be understood before any deployment.

## Radiative Forcing and Stratospheric Heating

Sulfate aerosols do two radiative things at once. At the surface, they scatter incoming shortwave solar radiation back to space, producing net cooling. In the lower stratosphere, they absorb near-infrared solar radiation and outgoing longwave radiation from the Earth below, producing localized warming at around 20 km altitude (Heckendorn et al., 2009). Heckendorn et al. (2009) simulated continuous  $\text{SO}_2$  injection of 1 to 10 Mt per year into the tropical lower stratosphere using a 3D chemistry-climate model and found that these aerosols grow to particle sizes larger than those observed after volcanic eruptions. Volcanic injections are brief, so the supply of fresh sulfuric acid vapor drops off quickly. Continuous injection keeps a steady supply of small fresh particles in the air, and they keep colliding and sticking together into larger ones, which sediment faster, depositing heat into the tropical cold-point tropopause. Heckendorn et al. (2009) found that warming the tropopause increases the amount of water vapor entering the stratosphere, because warmer air carries more moisture. More stratospheric water vapor then accelerates the hydroxyl radical ( $\text{HO}_x$ )-catalyzed ozone-destruction cycles, adding a depletion pathway.

Tilmes et al. (2022) quantified the stratospheric heating across six Earth system models in the GeoMIP G6sulfur experiment, where SAI was applied to keep surface warming at SSP2-4.5 levels despite SSP5-8.5 greenhouse gas concentrations. Tropical lower stratospheric

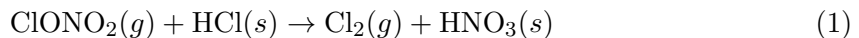
temperature increases ranged from 5 to 13 K across the six models. A steeper temperature difference between the warm tropics and the cold poles produces stronger westerly winds around Antarctica (Richter et al., 2017). Richter et al. (2017) confirmed this using CESM1(WACCM), running simulations with SO<sub>2</sub> injections at five different latitudes. In every case, regardless of where the SO<sub>2</sub> was injected, the tropical lower stratosphere warmed and the polar jets strengthened. Richter et al. (2017) also found that large equatorial injections disrupt the quasi-biennial oscillation (QBO), a natural roughly 28-month cycle of equatorial stratospheric winds. In a simulation with a 12 Tg/year equatorial injection, the QBO locked into a persistent westerly phase lasting more than 3.5 years. The easterly phase of the QBO normally lets large atmospheric waves travel upward into the polar stratosphere, where they deposit energy that warms and weakens the vortex. Without that phase, the vortex gets colder and more stable than the direct temperature gradient effect alone would predict.

## PSC Formation and Ozone Destruction

A stronger, colder polar vortex contains air better because it is isolated from warmer midlatitude air during the Southern Hemisphere winter, allowing temperatures to drop below what they would in a well-mixed atmosphere. Solomon (1999) described the vortex as providing two necessary conditions for severe ozone loss: cold temperatures that drive polar stratospheric cloud (PSC) formation, and isolation that keeps the processed air long enough for spring sunlight to trigger ozone destruction.

PSCs form in three compositions depending on temperature. Below 195 K, liquid sulfate aerosols grow into supercooled droplets composed of H<sub>2</sub>SO<sub>4</sub>, HNO<sub>3</sub>, and H<sub>2</sub>O. Below 191 K, solid nitric acid trihydrate particles can nucleate. Below 185 K, water ice clouds form (Solomon, 1999). SAI increases the amount of sulfate aerosol already floating in the stratosphere, so more liquid particles are available to grow into PSCs when temperatures drop low enough. Tilmes et al. (2008) used an empirical relationship between aerosol surface area density and chlorine activation to estimate that an injection large enough to compensate for CO<sub>2</sub> doubling would delay Antarctic ozone recovery by 30 to 70 years.

In the stratosphere away from the poles, most chlorine is sequestered in hydrochloric acid (HCl) and chlorine nitrate (ClONO<sub>2</sub>). On PSC surfaces, the heterogeneous reaction



proceeds rapidly (Solomon, 1999). Cl<sub>2</sub> accumulates in the isolated vortex air through winter, then photolyzes into chlorine radical atoms when sunlight returns in September. Each Cl radical then enters the catalytic ozone destruction cycle:



Each chlorine atom repeats the cycle many times before being deactivated (Solomon, 1999). The HNO<sub>3</sub> product of the heterogeneous reaction stays in the solid phase and can sediment

out of the stratosphere in a process called denitrification, removing the  $\text{NO}_x$  species that would otherwise deactivate  $\text{ClO}$  back into  $\text{ClONO}_2$ , extending the ozone destruction window into spring (Solomon, 1999). Pitari et al. (2014) confirmed across multiple GeoMIP models that ozone column loss tracks aerosol surface area, consistent with PSC chemistry. Tilmes et al. (2021) found that the hole deepens sharply in the first ten years after injection begins.

## Monitoring the System

Detecting and attributing ozone changes under SAI requires instruments capable of capturing relevant variables at appropriate altitudes during polar winter, when most remote sensing techniques fail. The primary tool is the Microwave Limb Sounder (MLS) on NASA’s Aura satellite. Waters et al. (2006) described MLS as an instrument that measures microwave radiation emitted by the atmosphere at its edge, scanning across a range of altitudes to build vertical profiles up to roughly 90 km, giving better vertical resolution than directly-downward-viewing instruments and allowing measurements through PSC and aerosol layers that block shorter-wavelength sensors.

The simultaneous  $\text{HCl}$  and  $\text{ClO}$  measurements are what make MLS useful.  $\text{HCl}$  is the dominant chlorine reservoir,  $\text{ClO}$  is the dominant reactive species that destroys ozone, and the transition from  $\text{HCl}$  to  $\text{ClO}$  is the direct chemical fingerprint of PSC activation. Santee et al. (2005) provided the first demonstration using MLS data from the 2004 Antarctic winter: mid-August maps at 20 km altitude showed a region over Antarctica where  $\text{HCl}$  was depleted and  $\text{ClO}$  was elevated, co-located with the coldest vortex temperatures. Ozone dropped throughout September as  $\text{ClO}$  catalyzed its destruction.

Strahan and Douglass (2018) used MLS data from 2005 to 2016 to track the long-term trend. Under ozone-depleted conditions inside the Antarctic vortex, inorganic chlorine ( $\text{Cl}_y$ ) is converted to  $\text{HCl}$ , so MLS  $\text{HCl}$  at the end of the ozone hole season is a direct measurement of total stratospheric chlorine. Over nine years, MLS showed  $\text{Cl}_y$  declined by 223 parts per trillion, confirming the Montreal Protocol is working. Under SAI,  $\text{Cl}_y$  would still decline due to the Protocol, but more of it would be converted to active  $\text{ClO}$  because extra aerosol surface area drives activation reactions, producing more ozone destruction per unit of chlorine than the current recovery trajectory assumes (Strahan & Douglass, 2018).

Nedoluha et al. (2016) tracked  $\text{ClO}$  in the Antarctic lower stratosphere for 20 years and found that September  $\text{ClO}$  levels correlate with 30 hPa temperature (correlation coefficient of  $-0.86$ ), indicating that colder temperatures produce more  $\text{ClO}$ . SAI strengthens the polar vortex, keeping it colder, so the same temperature-to- $\text{ClO}$  relationship this record captures is what SAI would produce as well.

## What the Models Show

Tilmes et al. (2022) in the GeoMIP G6sulfur multi-model analysis found that three models with dynamic stratospheric chemistry consistently showed Antarctic ozone column reduction, while three models with prescribed ozone showed inconsistent or negligible responses, concluding that models without coupled chemistry-dynamics cannot predict the ozone response to SAI. Only a dynamic model can capture that ozone changes alter the shortwave heating rate, which in turn alters temperature, which in turn alters vortex strength and PSC formation, which in turn alters chemistry.

Tilmes et al. (2021) found that the deepening of the ozone hole right after injection is

nonlinear. Higher injection rates lead to disproportionately larger losses because aerosol surface area grows faster than the amount of  $\text{SO}_2$ , as Heckendorn et al. (2009) showed. Models with predetermined aerosol sizes underestimate the surface area available for chlorine activation and the tropopause heating that adds a second ozone-loss pathway via  $\text{HO}_x$  chemistry.

The largest unresolved uncertainty is the QBO response. SAI warms the tropical stratosphere, strengthening the polar vortex, which cools the pole, which leads to more PSCs, which activate more chlorine, which destroys more ozone, which means less UV absorption, which cools the polar stratosphere further and strengthens the vortex again, a reinforcing feedback loop. Richter et al. (2017) found that locking the QBO into a persistent westerly phase removes the wave-driven warming that interrupts this loop. Niemeier and Schmidt (2020) found that two models respond differently to the same injection rate, so the extent to which this loop actually runs is unconstrained by current modeling.

## Future Research Directions

Bednarz et al. (2023) showed that extra-equatorial injections produce smaller polar ozone impacts than tropical injections, and Kravitz et al. (2019) found that spreading injections across multiple latitudes reduces polar vortex strengthening. No study has been specifically optimized to minimize Antarctic ozone depletion while still meeting surface temperature targets.

On alternative aerosols, Keith et al. (2016) proposed calcite ( $\text{CaCO}_3$ ), which could convert chlorine into stable  $\text{CaCl}_2$  that cannot activate on PSC surfaces. Vattioni et al. (2025) found in lab experiments that particle aging reduced reactive uptake, with a modeled net ozone change of only about  $-3\%$ . Alumina ( $\text{Al}_2\text{O}_3$ ) reduces stratospheric heating by up to  $70\%$  for the same radiative forcing (Vattioni et al., 2025), thereby reducing vortex strengthening.

## Conclusion

The chain from  $\text{SO}_2$  injection to ozone loss comprises three steps: aerosol heating strengthens the polar vortex (Richter et al., 2017), a colder and more isolated vortex extends the PSC season (Solomon, 1999), and PSC surfaces activate chlorine that destroys ozone (Tilmes et al., 2008). Recovery delay estimates range from 25 to over 70 years (Tilmes et al., 2021; Tilmes et al., 2008).

MLS observations confirm the Montreal Protocol is working (Strahan & Douglass, 2018; Santee et al., 2005). Whether SAI's climate benefits outweigh the ozone cost depends on emission scenarios and value judgments this review cannot resolve. What it establishes is that the ozone risk is real, mechanistically grounded, and not yet constrained by current modeling. Any serious evaluation of SAI has to account for it.

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